PROTEIN MALNUTRITION AND COGNITIVE STYLE IN HOODED RATS: AN ANTHROPOLOGICAL ANALOG

A Thesis

Presented to

The Faculty of

Western Washington State College

In Partial Fulfillment

Of the Requirements for the Degree

Master of Science

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by

Gary L. Michik

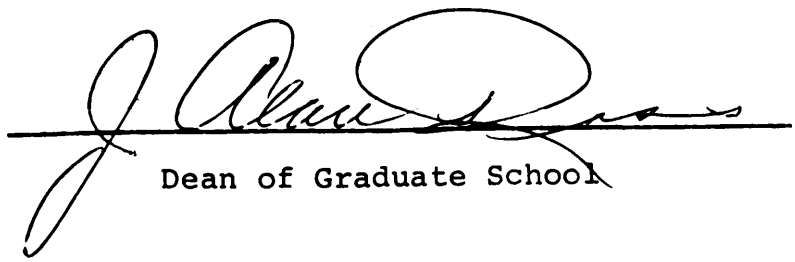
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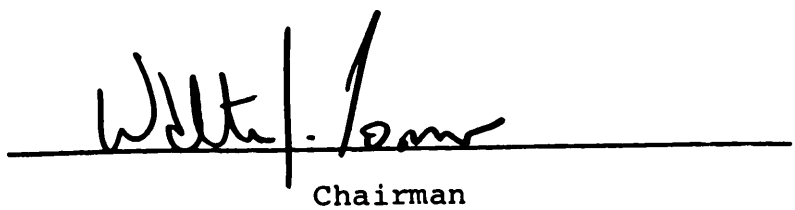
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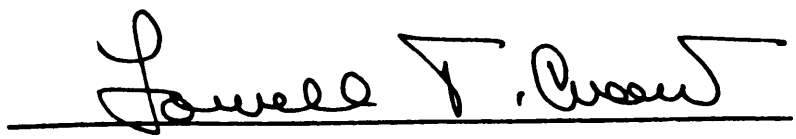


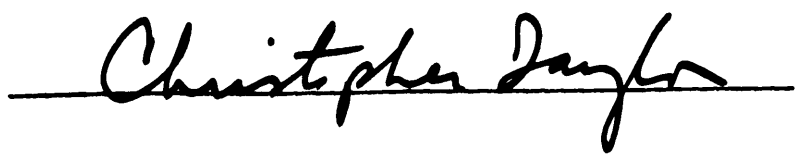
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The hypothesis of psychological differentiation is concerned with the communality among several areas of psychological functioning. The salient dimension for this hypothesis is the degree to which an individual's psychological system is differentiated (the degree to which the component parts are articulated). Empirical evidence demonstrating a similar degree of articulation among various areas of psychological functioning (e.g., perception, structuring and articulation of experience, articulation of body concept, sense of separate identity, analytical intellectual functioning) in humans is presented by Witkin, Dyk, Faterson, Goodenough, & Karp (1962). Thus, the degree of articulation in any one area may be considered an index of the general degree of differentiation of the entire system. This general degree of differentiation is defined as cognitive style.

Measures of analytical functioning in perception relating to the field dependence-independence dimension of perceptual style have provided important indexes of cognitive style. These tests measure a person's ability to separate an item from the field or context of which it is a part. A more field independent perceptual style is characterized by a greater

tendency to experience objects as discrete from their contexts and indicates a greater degree of psychological differentiation. A more field dependent perceptual style is characterized by a greater tendency to experience objects in a relatively global fashion, conforming to the influence of the field or context, and indicates a lesser degree of psychological differentiation (Witkin et al., 1962).

Empirical studies have indicated that males have a higher degree of psychological differentiation than females and that spatial ability has a high positive correlation with degree of differentiation (Witkin et al., 1962; Dawson, 1967a, b). Thus, a higher degree of psychological differentiation is sometimes called a more "masculine" cognitive style and a lower degree is sometimes called a more "feminine" cognitive style. In view of the consistant findings that females have a higher verbal/performance intelligence ratio than males (MacFarlane-Smith, 1964), Dawson (1972) has argued that this ratio is a valid measure of cognitive style. In the same paper, Dawson also argued that the consistant findings that male rats have higher spatial ability than female rats (Hubbert, 1915; Tyron, 1931; McNemar & Stone, 1932; Cowley & Griesel, 1963; Barnes, Cunnold, Zimmermann, Simmons, Macleod, & Krock, 1966) and the consistant findings that female rats have higher activity levels than male rats (Hitchcock, 1925; Richter, 1933; Harris, 1964; Gray, Lean, & Keynes, 1969) reflect the cognitive style in rats as accurately as measures of perceptual style, spatial ability, and verbal/performance intelligence ratio reflect the

cognitive style in humans.

This sex difference in cognitive style prompts examination of the relationship between sex hormones and cognitive style. Masica, Money, & Levis (1969) suggested that "... sexual hormonal effects in utero may be involved in the development of human cognitional patterns; and, more specifically, there may be a favorable influence of androgenic hormones on the developing brain ..." (p. 42), while fetal androgen levels may be responsible for sex differences in neural differentiation.

An important aspect of the cognitive style framework is the effect of nutrition on the development or inhibition of psychological differentiation. Kwashiorkor, for example, is a disease caused by early protein malnutrition. The disease produces feminization and gynaecomastia (along with other clinical abnormalities) in males (Trowell, 1949). Gynaecomastia is characterized by enlarged, feminine-appearing breasts and is due to a debilitation of the liver's ability to inactivate the normal amount of estrogens found in males (Stuart-Mason, 1963). Thus, kwashiorkor decreases the ratio of androgens to estrogens.

In a study of factory workers in West Africa, Dawson (1966) found that males manifesting the gynaecomastia symptom of kwashiorkor had more feminine cognitive styles than controls. He concluded that the sex hormone abnormality was at least partially responsible. In a later study, Dawson, Chueng, Lau, & Yue (in press) were able to reverse the normal sex differences in cognitive style by neonatal administration of opposite-sex

hormones in castrated rats. Dawson et al. concluded that neonatal sex hormone levels have an influence on psychological differentiation such that a higher neonatal androgen/estrogen ratio tends to produce a more masculine cognitive style.

These studies suggest the hypothesis that protein malnutrition produces a reduction in androgen/estrogen ratio which, in turn, leads to a more feminine cognitive style. Dawson (1969) has proposed a bio-social conceptual system which provides a theoretical framework for testing this hypothesis by means of cross-cultural research. In terms of this system, the degree of psychological differentiation is the result of an interaction between biological and cultural adaptations to the biological environment. The amount of protein available for consumption is considered part of that biological environment. While Dawson felt that other environmentally conditioned factors (e.g., modes of economic production, physical configuration of the environment, disease) were important determinants of cognitive style, he considered that dietary protein level would be a distinguishable factor in comparisons between societies with subsistance level economies because the modal diets could be expected to be close to the minimal requirements for survival.

Some support for the hypothesis that protein malnutrition produces a more feminine cognitive style is provided by a study of the Temne and Mende societies in the Sierra Leone area of West Africa (Dawson, 1967a, b). Dawson found that the Temne had more field dependent perceptual styles and

lower dietary protein levels than the Mende. A later study comparing samples from eight subsistance-level societies also provides evidence of the influence of dietary protein level on cognitive style. Berry (1971) found a high positive correlation between availability of protein for consumption and a more field independent perceptual style.

In both of these cross-cultural studies the index of cognitive style was found to be related to various other factors (e.g., harsh socialization practices, maternal dominance in the family, modes of economic production) as well as dietary protein level. Werner and Muralidharan (1970) reported a study of malnutrition in New Delhi nursery school children in which an attempt was made to control some of these other factors. They compared a group of 24 malnourished children with a control group of 16 children (from the same school and matched for parents' socio-economic status) who were receiving adequate diets. Werner and Muralidharan reported that the malnourished children were lower than controls in spatial ability and numerical ability and had more field dependent perceptual styles. Since the two groups were not significantly different on a test of verbal ability and numerical ability is an index of the performance factor of intelligence, I conclude that the malnourished group had a higher verbal/performance intelligence ratio. Thus, all these measures indicate that the malnourished group had more feminine cognitive styles than the control group, with the cultural factors at least partially controlled.

There have been a few studies of the effects of malnutrition on spatial ability and activity level in rats. As stated previously, rats with more masculine cognitive styles have higher spatial ability and lower activity levels than rats with more feminine cognitive styles. The reduction in spatial ability, predicted by the hypothesis that malnutrition produces a more feminine cognitive style, was found by Penner (1966) and Cowley & Griesel (1966). The findings for activity level are unclear. Guthrie (1968) and Livitsky & Barnes (1972) reported the hypothesized increase in activity level with malnutrition. However, Cowley & Griesel (1966) reported a decrease in the activity level of their malnourished animals. It is perhaps important to note that Guthrie, Penner, and Livitsky & Barnes subjected their experimental animals to early malnutrition followed by rehabilitation diets before testing, while Cowley & Griesel maintained their animals on an inadequate diet throughout their experiment.

Statement of Problem

The purpose of this experiment was to examine the effects of protein malnutrition on spatial ability and activity level in hooded rats. It was expected that protein-malnourished animals would have higher activity levels and lower spatial ability than controls, indicating that protein malnutrition produces a more feminine cognitive style. The independent variable in the present experiment (dietary protein level) was manipulated in a parent population before the actual

subjects measured on the dependent variables were conceived.

In view of the evidence indicating that sex hormones are related to protein malnutrition (Dawson, 1966; Stuart-Mason, 1963), spatial ability, and activity level (Dawson et al., in press), the weight of the sex glands was measured for each animal. Total body weight and weight of adrenal glands were also measured for each animal.

Method

Subjects

Twenty-two female and eight male hooded rats (Long-Evans strain, Western Washington State College rat colony) were bred, when eight months old, to produce the low protein experimental groups. The animals were maintained on an ad lib 8% protein diet (Nutritional Biochemicals Company, 1972 -- "Low 8% Protein Diet"). Ten days following initial exposure to the diet, the animals were allowed to breed for ten days. A total of 97 pups were born in 14 litters, 63 of which died before reaching sexual maturity. The pups were weaned when they were 21 days old and maintained on the same 8% diet as their parents throughout the experiment. All animals were housed individually after weaning. The 34 surviving experimental animals were separated by sex, forming a group of 16 males and a group of 18 females. Ten males were randomly selected and formed a group designated Experimental Male. Ten females were randomly selected and formed a group designated Experimental Female. These experimental groups were assumed to be random samples

from a hypothetical population of hooded rats surviving and reproducing on an 8% diet.

Six female and two male hooded rats (eight months old and from the same population as the parent animals used to produce the experimental groups) were used to produce the normal protein control groups. The same procedures were followed to produce these normal groups as were used to produce the experimental groups, except a 27% protein diet (Nutritional Biochemicals Company, 1972 -- "Normal Protein Test Diet") was used in place of the experimental 8% diet. A total of 13 pups were born in two litters, all of which lived to reach sexual maturity. These 13 control animals were separated by sex, forming a group of nine males and a group of four females. The male group was designated Control Male and the female group was designated Control Female. These control groups were assumed to be random samples from a hypothetical population of hooded rats surviving and reproducing on a 27% diet.

Activity Level

When the animals were approximately 65 days old, open field activity level was measured. The apparatus consisted of a wooden box 75 cm. square and ten cm. high with a wire mesh top. The sides were painted black and the floor was painted white with 36 12.5 cm. squares outlined in black. Each animal was placed in the middle of the box, once a day for five consecutive days, and observed for eight minutes. The dependent variable score for each subject was the sum

of lines crossed during all five sessions.

Spatial Ability

Hebb-Williams maze. When the animals were approximately 70 days old they were placed on a 12-h water deprivation cycle (11.5 h deprivation and 0.5 h ad lib drinking) for two days. Then the rats began the Hebb-Williams Closed Field Test as modified by Rabinovitch and Rosvold (1951).

During the adaptation phase, the rats received two 0.5-h sessions per day during the ad lib drinking period of their 12-h cycle (receiving water in the maze only). The adaptation phase was continued for six days. After the adaptation phase was completed, the animals began the preliminary training phase. Each rat was placed in the start box and allowed to go to the goal box and drink for 20 sec. before it was returned to the start box. Each animal spent 10 min. in the maze during each session with two sessions per day (receiving water only in the maze). The preliminary training phase was continued for eight days. Since none of the animals reached the criterion (as set up by Rabinovitch and Rosvold) for this phase, the Hebb-Williams test phase was not conducted.

Sunburst maze. Following the Hebb-Williams preliminary training phase, the animals were placed on a 24-h water deprivation cycle (23.5 h deprivation and 0.5 h ad lib drinking) for four days. At this time the animals were approximately 90 days old and began preliminary training on a modified version of the Tolman Sunburst Maze (Tolman, Ritchie, & Kalish, 1946).

The preliminary training problem is shown in Fig. 1. The

apparatus was constructed from unpainted plywood (1.69 cm. thick) and consisted of a circular disk 50 cm. in diameter and several paths 7.5 cm. wide. The whole maze was supported by a number of props which raised it about 60 cm. above the floor. Path AB was 30 cm. long and was used as the starting path. Paths CD, DE, and EF were all 45 cm. in length, while path FG was 150 cm. long. A goal box (30 cm. X 15 cm. X 15 cm.) was located at the end of path FG and had a water bottle mounted in it. A 15-watt light was located at H and provided the only illumination in the room. Fastened to the sides of path CD were two pieces of plywood (37.5 cm. high and 45 cm. long) which created an alley on that path.

Insert Figure 1 about here

On the first day of preliminary training the rats were given three trials. On the first trial they were placed in the goal box and allowed to drink for five min. On the second trial they were put in the middle of path FG and allowed to run to the goal box and drink for one min. On the third trial they were put at F and allowed to run to the goal box and drink for one min.

On the second day the animals were given three more trials. On the first trial they ran from F to the goal box and drank for one min. On the second and third trials they were placed in the alley on path CD and forced to run path

DE to the goal box where they drank for one min.

On the third day they were again given three trials. On the first trial they were forced to run out of the alley on path CD to the goal box where they drank for one min. On the second and third trials they were placed at A and allowed to explore the disk and run through the alley to the goal box where they drank for one min.

On the fourth day they were given three trials starting from A in the same manner as the last two trials on the third day. This completed the preliminary training. Each rat had run five times from A to the goal box at this time. On each of the four days the rats were allowed to drink for ten min. in their home cages after all the animals had completed their trials for the day.

On the fifth day one test trial was given. The apparatus was changed to that represented in Fig. 2. It consisted of the same starting path, disk, alley on path CD, and light at H. But the goal box and paths DE, EF, and FG were removed and a block was placed at the end of the alley on path CD. Then 18 unpainted plywood paths (150 cm. long and 7.5 cm. wide) were placed around the disk (see Fig. 2). These paths began at a point approximately 110 degrees to the right of path CD and radiated counterclockwise to a point approximately 110 degrees to the left of path CD. The paths were supported on props, raising them to the height of the rest of the maze, and could be rotated independently of the disk. For purposes of scoring, the paths were numbered counterclockwise from one

to 18 beginning with the first path to the right of B.

Insert Figure 2 about here

For the test trial each rat was placed at A, allowed to run into the blocked alley on CD, to return out of the alley and to explore the disk and the paths radiating from it. The rat's trial ended when it chose one of the paths by running out at least 30 cm. on it. The disk and the paths were rotated independently after each rat's trial to prevent any possibility of "tracking". The dependent variable score for each subject was the number of the path it chose on the test trial.

The animals were first trained to obtain water reward on the training maze shown in Fig. 1. Then that maze was replaced by the sunburst pattern shown in Fig. 2. Rats with high spatial ability were expected to choose the paths leading toward where the goal box had been (indicating that they had learned where the goal box was located), while rats with low spatial ability were expected to choose paths to the left of the alley on path CD (indicating that they had merely learned a series of responses which would bring them to the goal box) (Tolman et al., 1946). Tolman et al. obtained significant results with this test, although Gentry, Brown & Lee (1948) failed to replicate the origional findings. Ritchie (1948) did, however, find a significant side preference for the goal box, and Dawson (1972 was able to detect the sex difference

in spatial ability with this test. Gleitman (1955) argued that the problem with the work of Gentry et al. was due to their use of experimentally naive animals. The initial training of the rats in this study in the Hebb-Williams maze can be considered to have increased their test sophistication. Following completion of testing on this maze the animals were given ad lib water.

Physiological Measures

When the rats were approximately 98 days old they were sacrificed with chloroform. Following death, each animal's body was weighed on an Ohaus triple-beam balance and the adrenal and sex glands were removed and weighed on an electronic balance (Sartorius model #2743). These weights constituted the dependent variable scores for the physiological measures.

Results[1]

Behavioral Measures

Activity level. The results of the open field activity level measures are summarized in Table 1.

Insert Table 1 about here

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[1]The significances of differences among group means was tested by simple randomized analysis of variance (Lindquist, 1953). The significance of differences between individual group mean scores was tested by the Scheffé procedure (Winer, 1962). The alpha level for all statistical tests was 0.05.

Analysis of the variance of these data indicated that the differences among group mean scores were significant (F=16.13, df 3/29). The difference between the means of the two experimental groups was significant (S=8.29, df 1/29). The two control group means were not, however, significantly different (S=1.52, df 1/29). The results of the significance tests of differences between control group means and experimental group means are presented in Table 2.

Insert Table 2 about here

Spatial ability. The results of the test trial on the sunburst maze are summarized in Table 3.

Insert Table 3 about here

Analysis of the variance of these data indicated that the differences among group means were significant (F=5.45, df 3/29). The control male mean was significantly different from the control female mean (S=2.48, df 1/29), the experimental male mean (S=4.77, df 1/29), and the experimental female mean (S=8.09, df 1/29).

Physiological Measures

Body weight. The data from the measure of body weight are summarized in Table 4.

Insert Table 4 about here

Analysis of the variance of these data indicated that the differences among group means were significant (F=161.3, df 3/29). The difference between the means of the two experimental groups was not significant (S=2.68, df 1/29). The control male mean was, however, significantly higher than the control female mean (S=9.17, df 1/29). The results of the significance tests of differences between control group means and experimental group means are presented in Table 5.

Insert Table 5 about here

Weight of adrenals. The data from the measure of the weight of adrenal glands are summarized in Table 6.

Insert Table 6 about here

Analysis of the variance of these data indicated that the differences among group means were significant (F=60.98, df 3/29). The difference between the means of the two experimental groups was not significant (S=0.81, df 1/29) and the difference between the means of the two control groups was not significant (S=3.25, df 1/29). The results of the significance tests of differences between control group means and experimental group means are presented in Table 7.

Insert Table 7 about here

The ratio of weight of adrenal glands/body weight was computed for each animal. These ratios are summarized in Table 8.

Insert Table 8 about here

Analysis of the variance of these ratios indicated that the differences among group mean scores were significant (F=17.26, df 3/29). The experimental male mean was significantly lower than the experimental female mean (S=4.23, df 1/29) and the control male mean was significantly lower than the control female mean (S=6.32, df 1/29). The results of the significance tests of differences between control group means and experimental group means are presented in Table 9.

Insert Table 9 about here

Weight of sex glands. The data from the measure of the weight of sex glands are summarized in Table 10.

Insert Table 10 about here

The mean weight of testes for control males was significantly higher than the mean for experimental males. (S=37.6, df 1/17). The mean weight of ovaries for control females was significantly higher than that for experimental females (S=9.47, df 1/12).

The ratio of weight of sex glands/body weight was computed for each animal. These ratios are summarized in Table 11.

Insert Table 11 about here

The control male mean was significantly lower than the experimental male mean (S=8.93, df 1/17). The control female mean was not significantly different from the experimental female mean (S=0.04, df 1/12).

Discussion

Behavioral Measures

Activity level. Malnourished males had lower activity levels than control males and malnourished females had higher activity levels than malnourished males. The fact that control females were not found to be significantly different in activity level from either control males or malnourished females was probably due to the small size of the control female group (N=4). Therefore it will be assumed that these differences were significant (i.e., control females and higher activity levels than control males and malnourished females). Thus, this study indicates that protein malnutrition produces a decrease in activity level in both males and females, that the normally higher activity level in females is maintained in the malnourished animals, and that protein malnutrition produces a greater decline in activity level in males than it does in females.

These results contradict the hypothesis that protein

malnutrition produces higher activity levels in rats and are in agreement with the findings of Cowley & Griesel (1966) showing that lower activity levels are produced. In both of the studies which examined the effects of protein malnutrition on activity level and obtained results indicating that protein malnutrition produces an increase in activity level (Guthrie, 1968; Livitsky & Barnes, 1972) the experimental animals received rehabilitation diets prior to testing. However, in this study, and in the study by Cowley & Griesel (1966), rehabilitation diets were not used and the opposite results were obtained. This suggests that a particular nutritional sequence (early malnutrition followed by rehabilitation on an adequate diet) is necessary to produce the increase in activity level which is predicted by the hypothesis that protein malnutrition produces a more feminine cognitive style.

Spatial ability. The control males had a mean of 6.2 on the test trial in the sunburst maze. This was almost directly toward the previous location of the goal box (path #6 led directly to where it had been) and indicates high spatial ability. The control females had a mean of 9.8, which is to the left of the training path, indicating that they had lower spatial ability than the control males. The malnourished males had a mean of 9.7, which is also to the left of the training path, indicating that they had lower spatial ability than the control males. Since the method used to measure spatial ability (the sunburst maze) can only indicate a dichotomous distinction between higher and lower

spatial ability, comparisons among the malnourished male, malnourished female, and control female group means were superflous because all three of these groups fell into the lower category.

These results support the findings of Cowley & Griesel (1966) and Penner (1966) showing that protein malnutrition produces lower spatial ability in rats. They are also consistant with the hypothesis that protein malnutrition produces a more feminine cognitive style.

Physiological Measures

The physiological measures were obtained primarily to examine their possible relationships to the behavioral measures. It was hoped that these measures would shed some light on the physiological mechanisms by which protein malnutrition affects the behavior of the animals.

Body weight. Control males were heavier than control females. There was, however, no relationship between sex and body weight in the malnourished animals. The mean body weights of both malnourished groups were less than either of the control group means.

These results indicate that protein malnutrition abolishes the normal sex difference in body weight as well as producing a reduction in weight in both sexes. Thus, the effect of protein malnutrition on body weight is more severe for males than it is for females. It is interesting to note that this greater weight deficit in males is accompanied by a greater change in activity level compared

to females and therefore may be generally related to sex differences in cognitive style. Further research will be necessary to determine the relationship between the amount of nutrients necessary to maintain body weight and cognitive style.

Weight of adrenal glands. There were no sex differences in weight of adrenal glands for either control or malnourished animals. The adrenal glands of the control animals were heavier than those of the malnourished animals for both sexes.

These results indicate that protein malnutrition produces lighter adrenal glands. Inspection of the data did not indicate any apparent relationships between weight of adrenal glands and activity levels or spatial ability.

It is reasonable to assume that hormonal action is dependent on relative concentration of the hormones rather than the absolute amount of the substances present. Since the malnourished animals were smaller than the control animals, their adrenal glands would not have had to secrete as great a volume of hormones to produce the same relative concentration. If the weight of the adrenal glands is an index of the volume of hormones they secrete, the lighter adrenal glands of the malnourished animals may have produced the same, or possibly a higher relative concentration of catecholamines compared to that produced by the heavier adrenal glands of the control animals.

It was, therefore, felt that the ratio of weight of adrenal glands to body weight might provide a better index

of the degree of adrenal influence on behavior. These ratios were computed and analyzed. Females had higher relative adrenal weights than males in both the control and the malnourished subjects. Malnourished females had higher relative adrenal weights than control females and malnourished males had higher relative adrenal weights than control males.

These results indicate that females had higher relative adrenal weights than males for both the malnourished and control groups. The results of the measures of activity level and spatial ability correlate with these findings suggesting the hypothesis that rats with higher relative adrenal weights have higher activity levels and lower spatial ability (or a more feminine cognitive style) than rats with lower relative adrenal weights. Dawson et al. (in press) have discussed the sex differences in cognitive style in terms of the superiority of male rats over female rats on tasks involving response inhibition such as tests of spatial ability, and have suggested that this greater capability of males to inhibit responses is also responsible for their lower activity levels. Assuming that relative catecholamine concentration is negatively correlated with ability to inhibit responses, the fact that females had lower spatial ability and higher activity levels than males could be due to higher relative catecholamine concentrations produced by the females' relatively heavier adrenal glands. This would explain the sex differences in cognitive style.

Malnourished animals, however, have higher relative

adrenal weights than controls and also have lower, rather than higher, activity levels. It would appear that while relative adrenal weight is related to sex differences in cognitive style in both malnourished and control animals, protein malnutrition changes the relationship between relative adrenal weight and volume of catecholamines secreted in both sexes.

Weight of sex glands. Control males had heavier testes than malnourished males and control females had heavier ovaries than malnourished females. This finding for males is consistant with the hypothesis that males with lighter testes are lower in spatial ability and lower in activity level than males with heavier testes. This finding for females is consistant with the hypothesis that females with heavier ovaries have higher activity levels than females with lighter ovaries. The relationship between ovary weights and spatial ability is not discussed since the sunburst maze did not differentiate between the two groups of females.

By the same logic employed in the discussion of the results of the measure of the weight of adrenal glands, it was felt that the ratio of weight of sex glands to body weight would provide a more valid index of gonadal influence on behavior. These ratios were computed and analyzed. Malnourished males had relatively heavier testes than control males. Malnourished females had relatively heavier ovaries than control females, but this difference was not significant. This lack of significance was probably due to the small size

of the control female group (N=4) and it is therefore assumed that protein malnutrition does produce relatively heavier ovaries in females.

The difference in relative weight of testes suggests that males with relatively lighter testes are higher in spatial ability and activity level than males with relatively heavier testes. In view of the evidence indicating a positive correlation between androgen level and spatial ability (Dawson et al., in press), these results are consistant with the hypothesis that the hypertrophied testes of the malnourished rats produced relatively smaller amounts of androgens and, consequently, lower spatial abiltiy. Dawson et al. also reported a negative correlation between androgen level and activity level, which was not found in this study. However, that study did not involve protein malnutrition and this study has indicated that protein malnutrition affects activity level by mechanisms other than sex hormone levels.

The difference in relative weight of ovaries suggests the hypothesis that females with relatively heavier ovaries have lower activity levels than females with relatively lighter ovaries. In view of the evidence indicating a positive correlation between estrogen level and activity level (Dawson et al., in press) these results are consistant with the hypothesis that hypertrophied ovaries, like hypertrophied testes, produce relatively smaller amounts of estrogens and, consequently, relatively lower activity levels. Apparently sex hormone levels are a more important determinant

of activity level in females than they are in males, in which adrenal activity is more important. This is probably related to the greater decrease in activity level in males compared to females with protein malnutrition.

Summary

The purpose of this study was to test the hypothesis that protein malnutrition produces a more feminine cognitive style in rats as defined by lower spatial ability and higher activity level. The results of the test of spatial ability support this hypothesis, but the results of the measure of activity level do not. Examination of the methods and results of this study and studies by Penner (1966), Cowley & Griesel (1966), Livitsky & Barnes (1972), and Guthrie (1968) led to the suggestion that a particular nutritional sequence during the development of the organism (early malnutrition followed by an adequate, rehabilitating diet) was necessary to produce the heightened activity level predicted by the hypothesis. The results of the physiological measures indicated that adrenal activity, as well as sex hormone levels, was involved with cognitive style and that this adrenal involvement was more important in males than in females in the determination of cognitive style. These results are consistant with the hypothesis that protein malnutrition (with rehabilitation) leads to a decrease in androgen/estrogen ratio and an increase in catecholamine level, which in turn, lead to a more feminine cognitive style.

The question remains as to whether the nutritional

sequence necessary to produce a more feminine cognitive style in rats also occurred in the anthropological studies cited in this paper. In the cross-cultural studies by Berry (1971) and Dawson (1967a, b), and in the study of nursery school children by Werner & Muralidharan (1970), the researchers did not collect data on the nutritional history of their subjects. Although Dawson (1966) also failed to collect such data in his study of factory workers in West Africa, their nutritional history can be inferred. The subjects found to have more feminine cognitive styles had all suffered from kwashiorkor when they were young. At the time Dawson studied them they were holding jobs in a factory so they had money to buy food and were eating well enough to stay healthy. Thus, at least for this study, the nutritional sequence necessary to produce a more feminine cognitive style in rats had occurred in the humans found to have more feminine cognitive styles.

This paper suggests that, if an organism is malnourished when young and does not subsequently receive an adequate, rehabilitating diet, the degree of articulation of one area of psychological functioning (e.g., spatial ability) will not be correlated with the degree of articulation of another area of psychological functioning (e.g., activity level), in conflict with what the hypothesis of psychological differentiation (Witkin et al., 1962) would predict. However, if the organism does receive an adequate, rehabilitating diet after early malnutrition, a similar

degree of articulation will be manifested in both areas of psychological functioning, supporting the differentiation hypothesis.

This conflict suggests that malnutrition affects different areas of psychological functioning differently while it is occurring, but that these differences in effect disappear after rehabilitation (when a roughly similar effect is manifested in all areas of psychological functioning). This could be explained by assuming that malnutrition produces a similar reduction in the degree of articulation of the neural substrates of all areas of psychological functioning, but that some of these areas (e.g., activity level) are more dependent on continuous, adequate replenishment of critical nutrients for their behavioral expression than others (e.g., spatial ability). While the degree of articulation of the neural substrates of all areas of psychological functioning would be similar (as predicted by the differentiation hypothesis), behavioral indexes of the degree of articulation of different areas of psychological functioning could indicate differences in degree of articulation. If the organism then received an adequate, rehabilitating diet, the actual degree of neural differentiation would be reflected in all areas of psychological functioning and would support the differentiation hypothesis.

References

Barnes, R. H., Cunnold, S. R., Zimmermann, R. R., Simmons, A., Macleod, R. B., & Krock, L. Influence of nutritional deprivations in early life on learning behavior of rats as measured by performance in a water maze. Journal of Nutrition, 1966, 89, 4.

Berry, J. W. Ecological and cultural factors in spatial perceptual development. Canadian Journal of Behavioral Science, 1971, 3(4), 324-336.

Cowley, J. J. & Griesel, R. D. The development of second generation low-protein rats. Journal of Genetic Psychology, 1963, 103, 233-242.

Cowley, J. J. & Griesel, R. D. The effect on growth and behavior of rehabilitating first and second generation low protein rats. Animal Behavior, 1966, 14, 506-517.

Dawson, J. L. M. Kwashiorkor, gynaecomastia, and feminization processes. Journal of Tropical Medicine and Hygiene, 1966, 69, 175-179.

Dawson, J. L. M. Cultural and physiological influences on spatial-perceptual processes in West Africa: Part I. International Journal of Psychology, 1967, 2, 115-128, (a).

Dawson, J. L. M. Cultural and physiological influences on spatial-perceptual processes in West Africa: Part II. International Journal of Psychology, 1967, 2, 171-185, (b).

Dawson, J. L. M. Research and theoretical bases of bio-social psychology: Inaugural lecture from the Chair of Psychology: University of Hong Kong. Supplement to the Hong Kong Gazette, 1969, 16, 1-10.

Dawson, J. L. M. Effects of sex hormones on cognitive style in rats and men. Journal of Behaviour Genetics, 1972, 2, 20-41.

Dawson, J. L. M., Chueng, Y. M., Lau, T. S., & Yue, F. The neonatal sex hormone reversal of cognitive skills in rats and men. (in press).

Gentry, G., Brown, W. L., & Lee, H. Spatial location in the learning of a Multiple T. Maze. Comparative Physiology, 1948, 41, 312-318.

Gleitman, H. Place learning without prior performance. Journal of Comparative and Physiological Psychology, 1955, 48, 77-79.

Gray, J. A., Lean, J., & Keynes, A. Infant androgen treatment and adult open-field behavior: Direction effects and effects of injections to siblings. Physiology and Behavior, 1969, 4, 177-181.

Guthrie, H. A. Undernutrition, the brain composition and behavior in rats: The effect of the timing and extent of undernutrition during early life on adult behavior, brain growth, and brain composition in rats. Dissertation Abstracts, 1968, 29(5-B), 1738.

Harris, G. W. Sex hormones, brain development and brain function. Endocrinology, 1964, 75 627-648.

Hitchcock, F. A. Studies in vigor V. The comparative activity of male and female albino rats. American Journal of Psychology, 1925, 75, 205-210.

Hubbert, H. B. The effect of age on habit formation in the albino rat. Behavior Monograph, 1915, 2, 1-55.

Lindquist, E. F. Design and analysis of experiments in psychology and education. Boston: Houghton Mifflin Company, 1953.

Livitsky, D. A. & Barnes, R. H. Nutritional and environmental interactions in the behavioral development of the rat: Long term effects. Science, 1972, 176, 66-71.

MacFarlane-Smith, I. Spatial ability. London: University of London Press Ltd., 1964.

McNemar, Q. & Stone, C. P. The set difference in rats on three learning tasks. Journal of Comparative Psychology, 1932, 14, 171-180.

Masica, D. N., Money, J., Ehrhardt, A. A., & Levis, V. G. I. Q., fetal sex hormones, and cognitive patterns: Studies in the testicular feminizing syndrom of androgen insensitivety. John Hopkins Medical Journal, 1969, 124, 1, 34-43.

Nutritional Biochemicals Company, Diets Manual, Cleveland, Ohio, 1972.

Penner, L. R. The effect of pre- and early post natal protein deficiency on maturation and intelligence in the albino rat. Dissertation Abstracts, 1966, 27(3-B), 988.

Rabinovitch, M. S. & Rosvold, H. E. A closed field intelligence test for rats. Canadian Journal of Psychology, 1951, 5, 122-128.

Richter, C. P. Activity cycles. Endocrinology, 1933, 17, 445.

Ritchie, B. F. Studies in spatial learning VI. Place orientation and direction orientation. Journal of Experimental Psychology, 1948, 38, 659-669.

Stuart-Mason, A. Health and hormones. London: Pelican, 1963.

Tolman, E. C., Ritchie, B. F., & Kalish, D. Studies in spatial learning. I. Orientation and the short-cut. Journal of Experimental Psychology, 1946, 36, 13-24.

Trowell, H. C. Medical aspects of inefficiency. In C. H. Northcott (Ed.), African Labour Efficiency Survey, London: H. M. S. O., 1949.

Tyron, B. C. Studies in individual differences in maze ability II. The determination of individual differences by age, weight, sex, and pigmentation. Journal of Comparative Physiology, 1931, 12, 1-22.

Werner, E. F. & Muralidharan, R. Nutrition, cognitive status, and achievement motivation of New Delhi nursery school children. Journal of Cross-Cultural Psychology, 1970, 1, 271-281.

Winer, B. J. Statistical principals in experimental design, New York: McGraw-Hill, 1962.

Witkin, H. A., Dyk, R. B., Faterson, H. F., Goodenough, D. E., & Karp, S. A. Psychological differentiation, New York: Wiley, 1962.

Figure 1

Apparatus Used in Preliminary Training

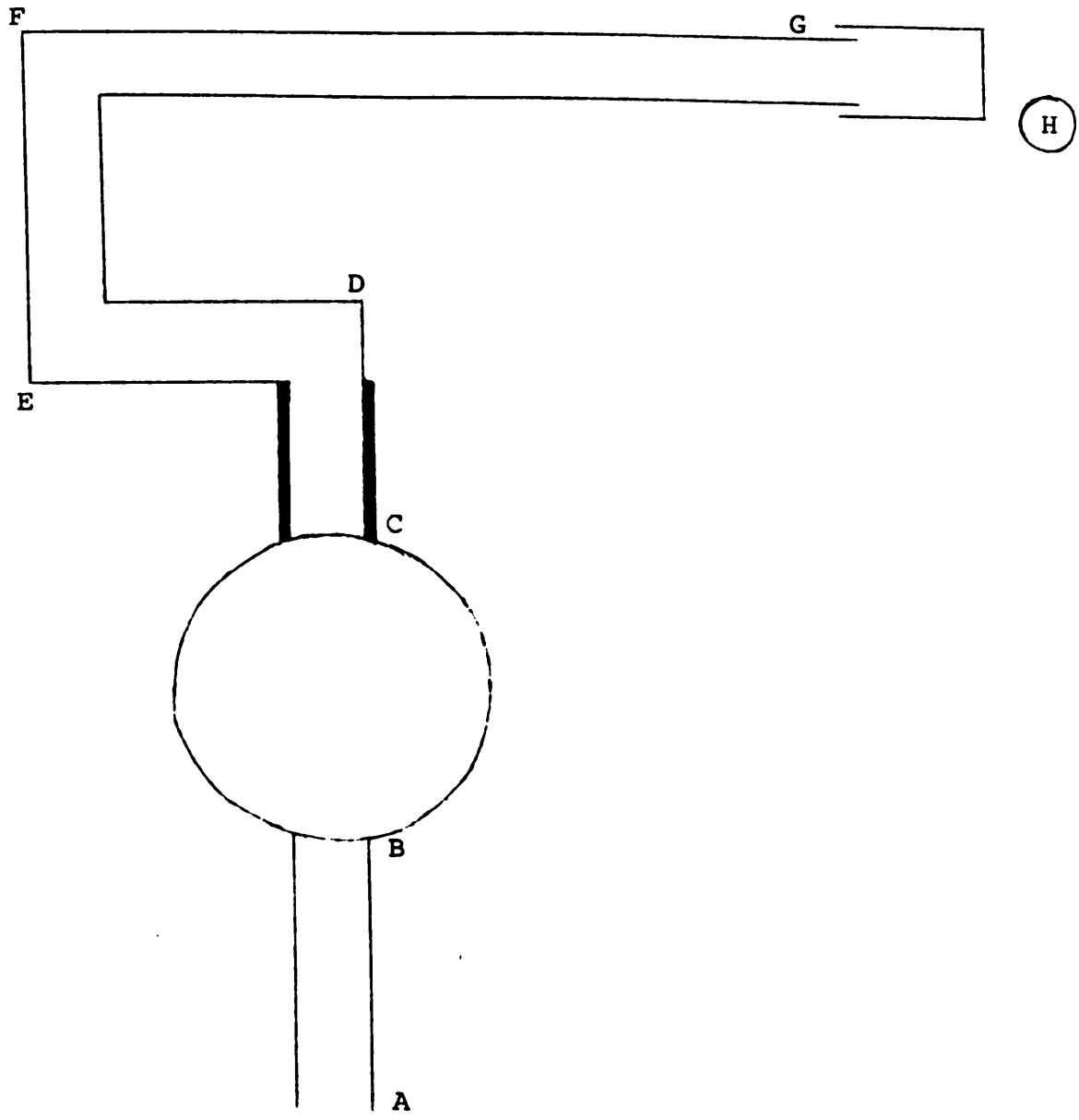


Figure 2

Apparatus Used in Test Trial

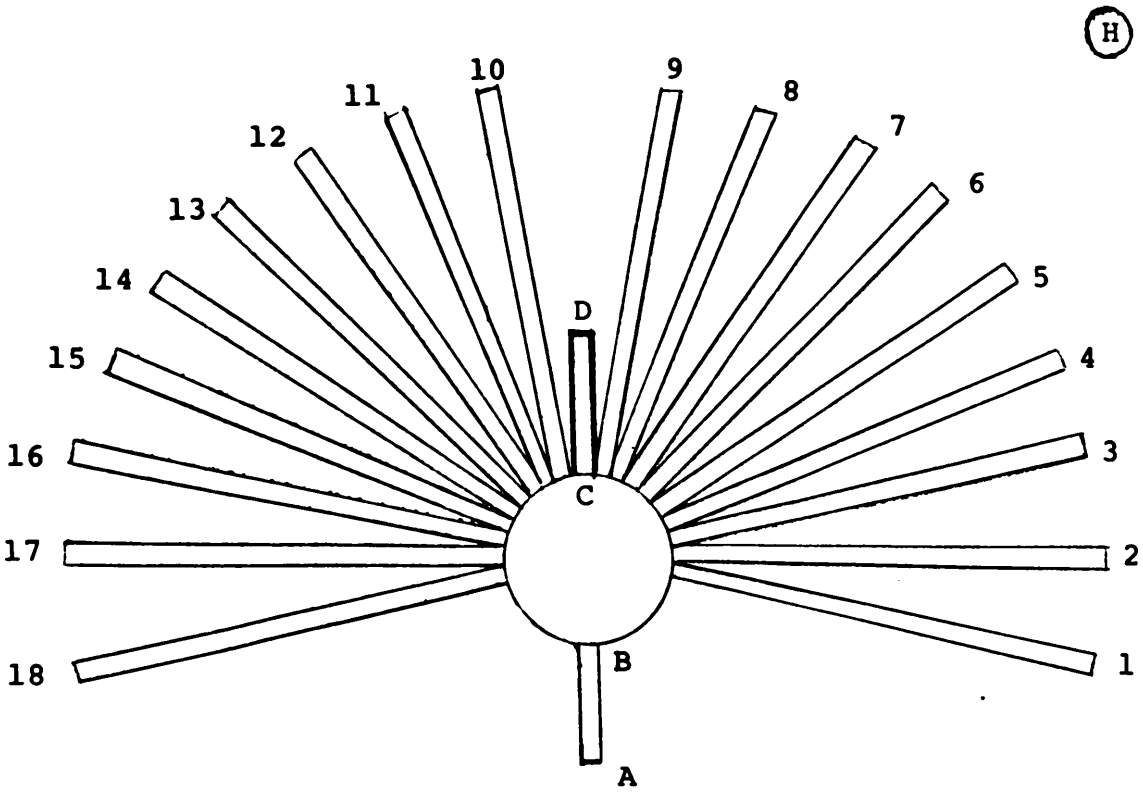


Table 1

Open Field Activity Level Data: Number of Lines Crossed

	Mean	Standard Deviation
Control Males	560	261
Control Females	619	56
Experimental Males	308	165
Experimental Females	550	277

Table 2

Significance Tests of Differences in Activity Level Between Control and Experimental Groups

Group	Control Male	Control Female
Experimental Male	S=8.48, df 1/29*	S=8.32, df 1/29*
Experimental Female	S=0.34, df 1/29	S=2.36, df 1/29

*Significant

Table 3

Sunburst Maze Data: Number of First Path Entered

	Mean	Standard Deviation
Control Males	6.2	3.048
Control Females	9.8	3.642
Experimental Males	9.7	3.467
Experimental Females	12.1	2.441

Table 4

Body Weight Data (grams)

	Mean	Standard Deviation
Control Males	289	31
Control Females	227	31
Experimental Males	105	10
Experimental Females	91	18

Table 5

Significance Tests of Differences in Body Weight Between Control and Experimental Groups

Group	Control Male	Control Female
Experimental Male	S=35.67, df 1/29*	S=18.41, df 1/29*
Experimental Female	S=38.28, df 1/29*	S=20.44, df 1/29*

*Significant

Table 6

Weight of Adrenal Glands (grams)

	Mean	Standard Deviation
Control Males	0.045	0.0054
Control Females	0.050	0.0093
Experimental Males	0.024	0.0031
Experimental Females	0.023	0.0032

Table 7

Significance Tests of Differences in Adrenal Weight Between Control and Experimental Groups

Group	Control Male	Control Female
Experimental Male	S=19.08, df 1/29*	S=18.13, df 1/29*
Experimental Female	S=19.87, df 1/29*	S=18.74, df 1/29*

*Significant

Table 8

Weight of Adrenal Glands/Body Weight

	Mean	Standard Deviation
Control Males	0.00016	0.0000173
Control Females	0.00022	0.0000223
Experimental Males	0.00023	0.0000224
Experimental Females	0.00026	0.0000480

Table 9

Significance Tests of Differences in Adrenal Weight/ Body Weight Between Control and Experimental Groups

Group	Control Male	Control Female
Experimental Male	S=9.59, df 1/29*	S=1.41, df 1/29
Experimental Female	S=12.33, df 1/29*	S=4.26, df 1/29*

*Significant

Table 10

Weight of Sex Glands (grams)

	Mean	Standard Deviation
Control Males	2.97	0.274
Experimental Males	1.58	0.361
Control Females	0.095	0.006
Experimental Females	0.039	0.023

Table 11

Weight of Sex Glands/Body Weight

	Mean	Standard Deviation
Control Males	0.010	0.00112
Experimental Males	0.015	0.02980
Control Females	0.00042	0.000045
Experimental Females	0.00046	0.000110